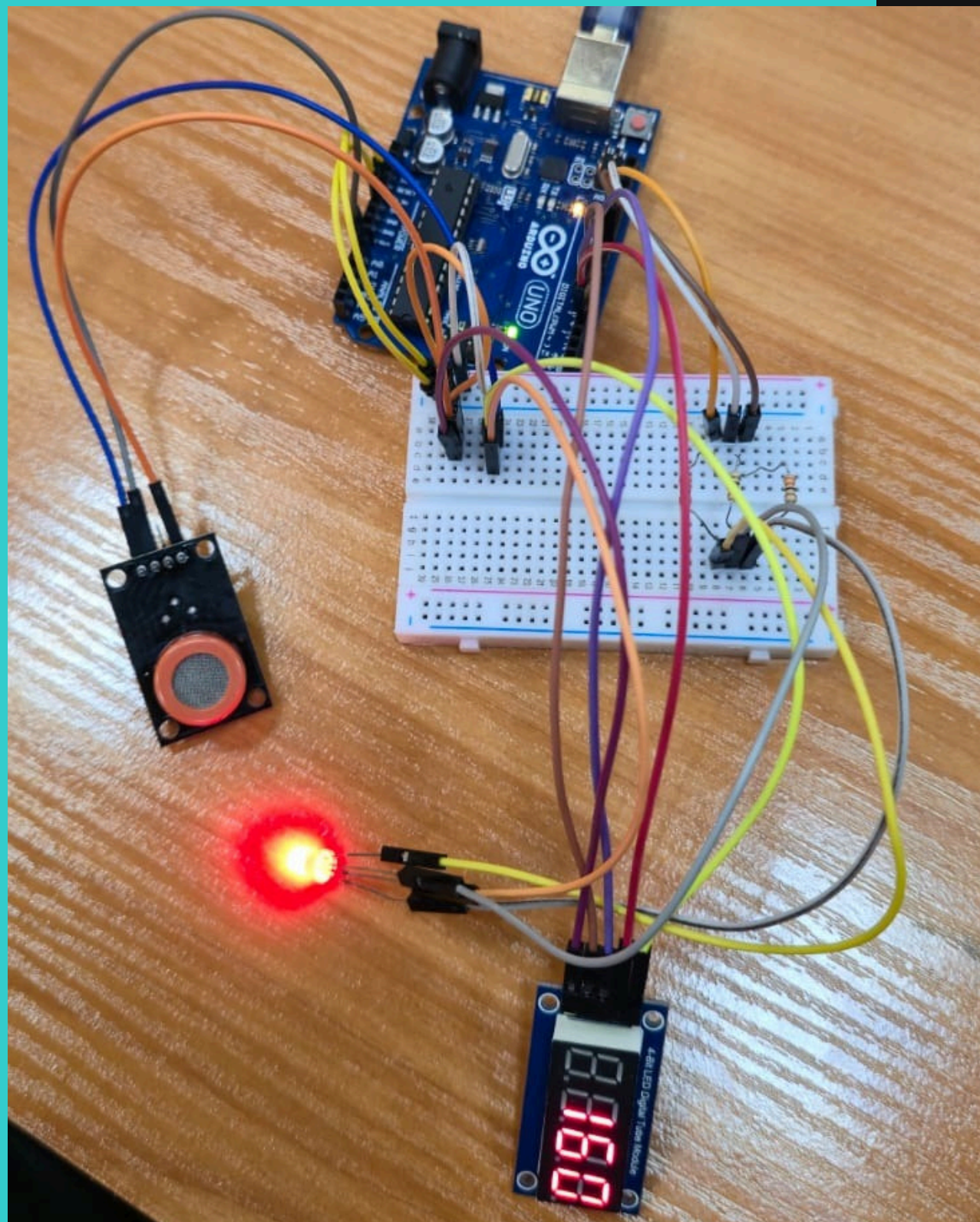




Breathalyzer

Electronic Circuits and Data Transmission in Medical Equipment

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Project Overview & Objectives

Goal: Convert analog gas signals into readable BAC metrics (promils).

Core Hardware:

- CPU: Arduino Uno.
- Sensor: MQ-3.
- Output: 4-Digit-7-Segment Display.

Key Innovation:

- Original Requirement: 5 separate LEDs.
- Our Design: Single RGB LED (PWM controlled).
- Benefit: Dynamic color mixing and circuit simplification.

Circuit architecture

Input:

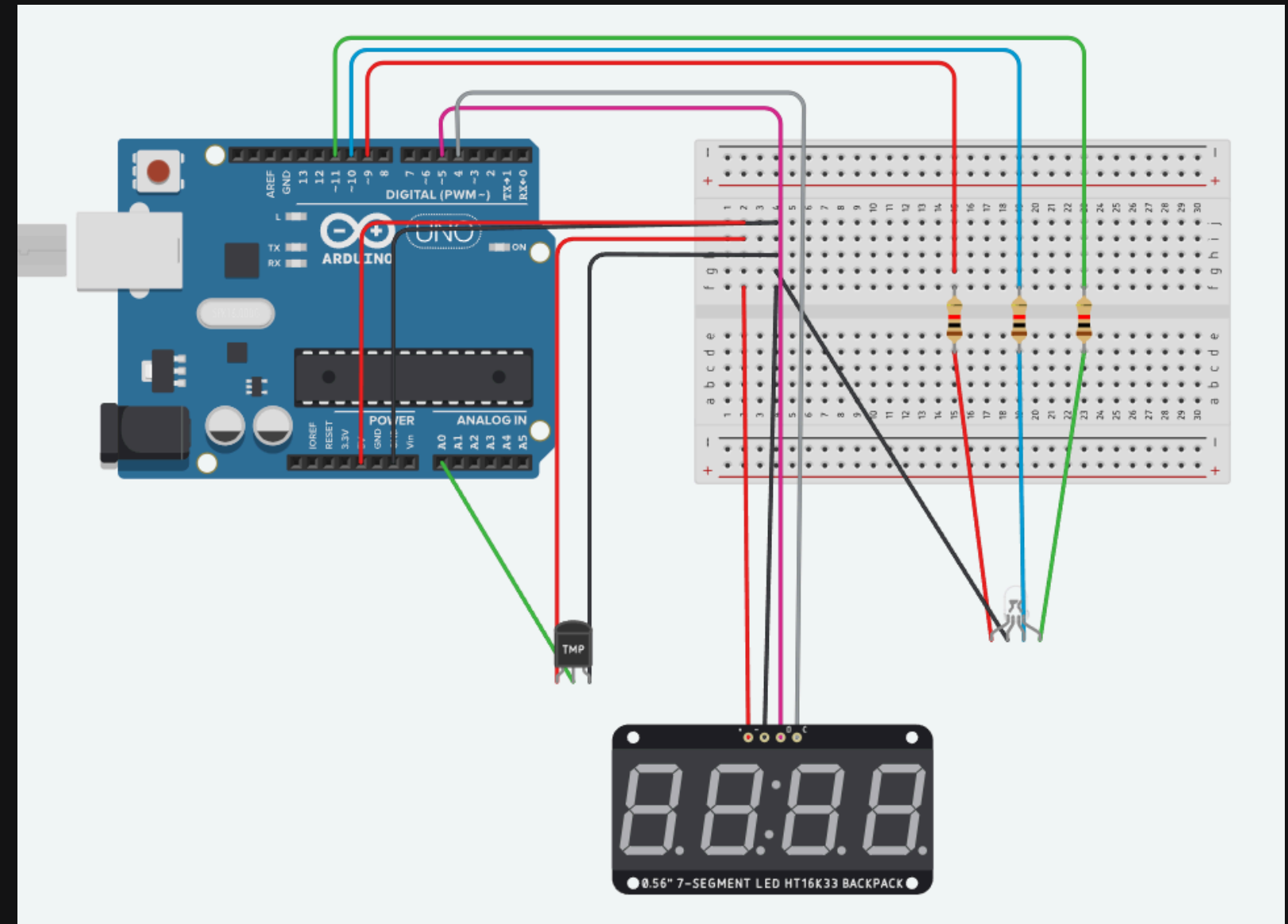
- MQ-3 Sensor connected to Analog Pin A0.

Display Logic:

- Drives the 4-digit display using only 3 Pins (4, 5, 6) instead of 12.

Visual Feedback:

- RGB LED connected to PWM Pins (9, 10, 11).
- Enables color mixing.



MQ-3 Sensor physics

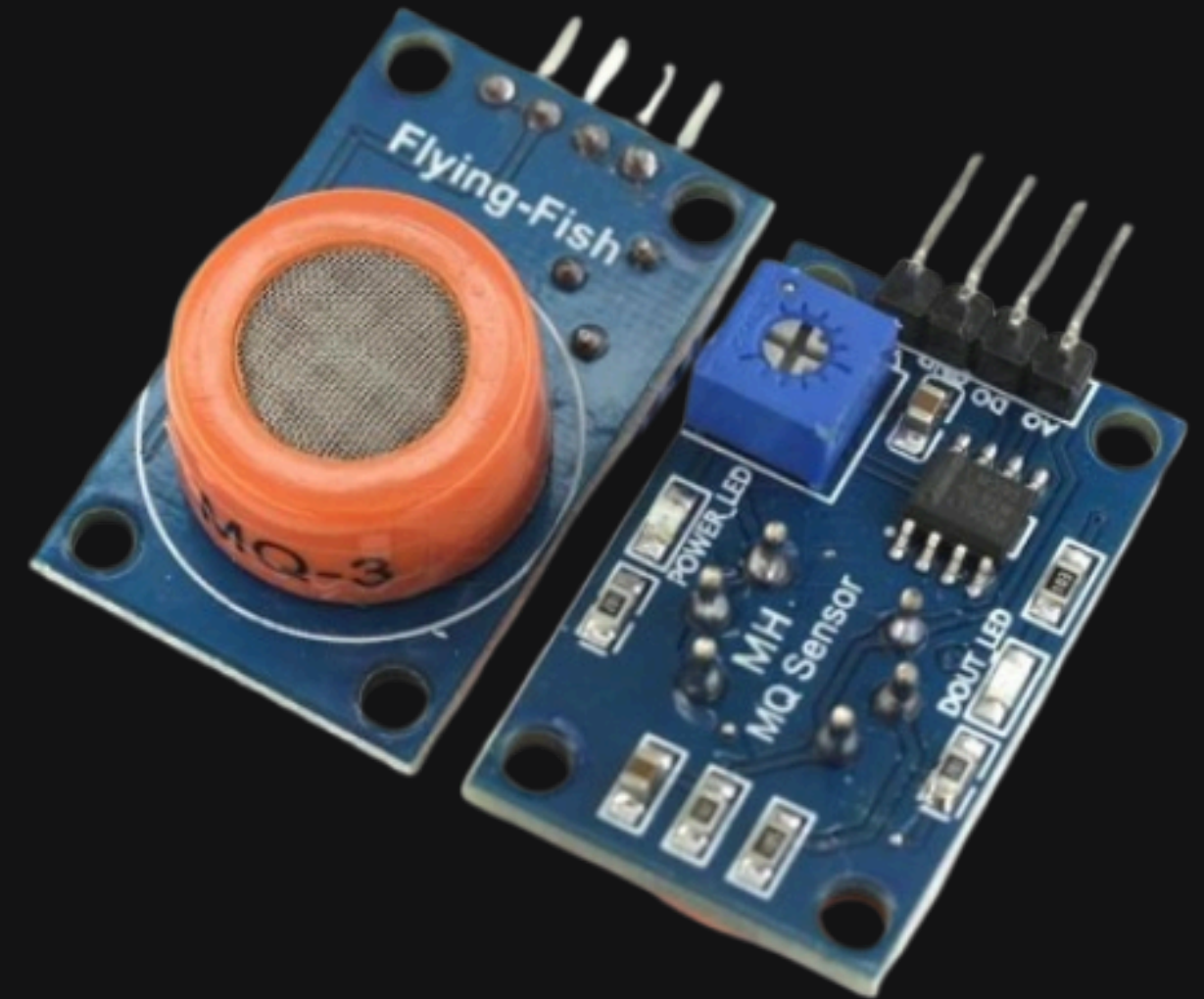
Component: Tin Dioxide (SnO_2) ceramic tube with internal heater.

Mechanism:

1. Clean Air: Oxygen adsorbs to surface $\rightarrow$ Traps electrons $\rightarrow$ High Resistance.
2. Alcohol Vapor: Ethanol reacts with oxygen $\rightarrow$ Releases electrons $\rightarrow$ Low Resistance.

Result:

- Physical chemical reaction is converted into an electrical resistance drop.



Mathematical model

From Resistance to Promils:

1. Voltage Divider Formula:
$$V_{out} = 5V \times \left(\frac{10k\Omega}{R_s + 10k\Omega} \right)$$

2. BAC Calculation (Power Law):
$$BAC = 0.2 \times \left(\frac{R_s}{R_0} \right)^{-1.9}$$

- R_0 : Baseline calibrated resistance.
- R_s : Current sensor resistance.

Thermal management

Power Consumption Analysis:

- Heating Element: MQ-3 requires heat to activate the chemical reaction.
- Heater Resistance: 33Ω .
- Power Calculation :

$$P = \frac{V^2}{R} = \frac{5^2}{33} \approx 760 \text{ mW}$$

- Current Draw: 150mA.
- Conclusion: The device operates within safe thermal limits (40-50 °C) without requiring external cooling.

Key Algorithms & Logic

1. Auto-Calibration Routine:

To ensure accuracy, the system calculates the baseline resistance by averaging 100 samples upon startup

2. Non-Blocking Timing:

Instead of using `delay()` we utilized `millis()`, allowing the display multiplexing to run continuously without interruption.

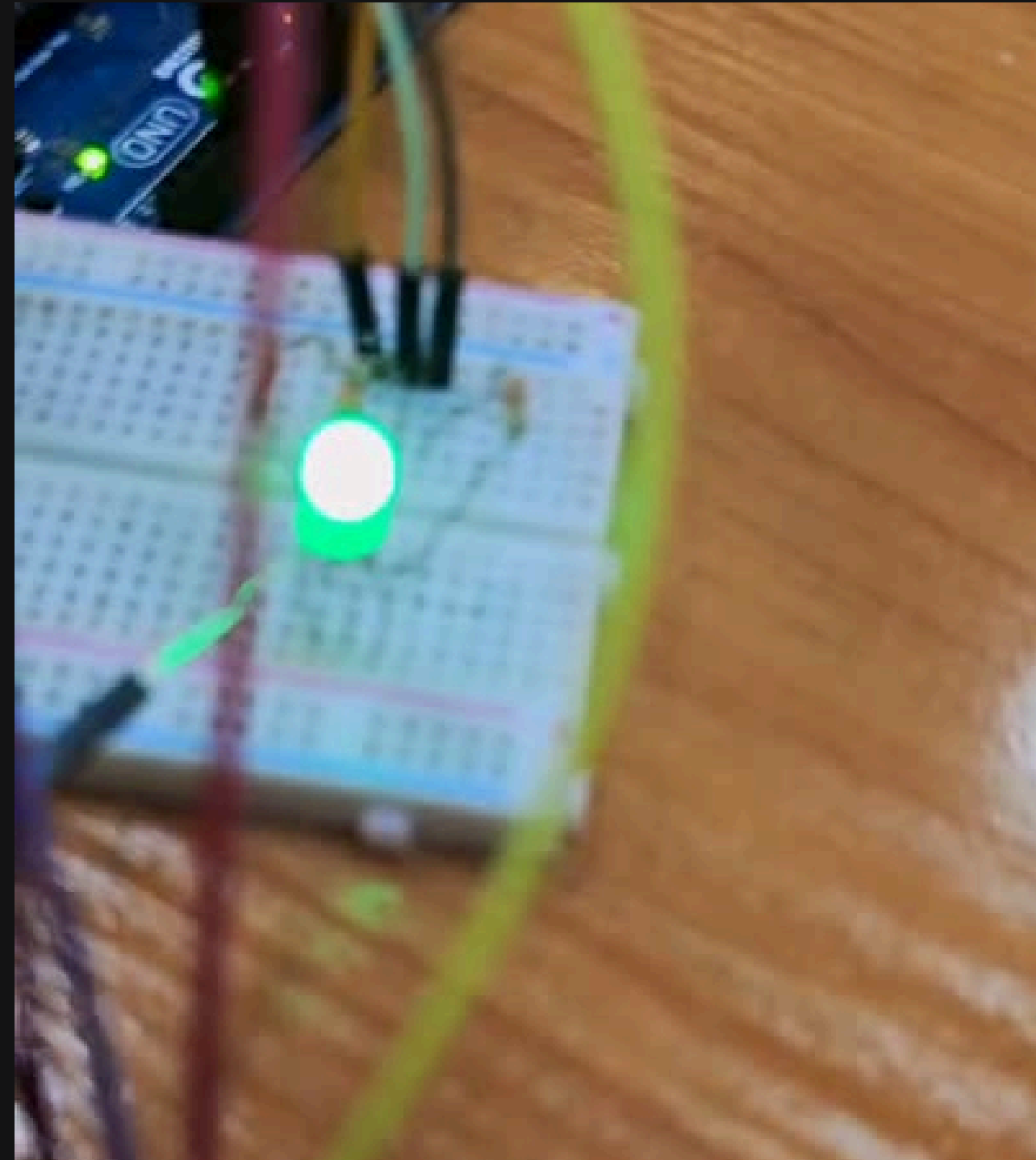
3. Common Anode RGB Control:

Since the LED is Common Anode, the logic is inverted ($255 - \text{value}$)

Experimental Results & Validation

- **Test Source:** We utilized an alcohol-based hand sanitizer to simulate a high-intoxication environment safely.
- **Threshold Adjustment:** During the testing phase, we fine-tuned the detection thresholds in the code to simplify the result visualization
- **Visual Confirmation:** The RGB LED successfully transitioned between the three colours. The 4-digit display updated in real-time.

Result Video



Conclusion

Achievements:

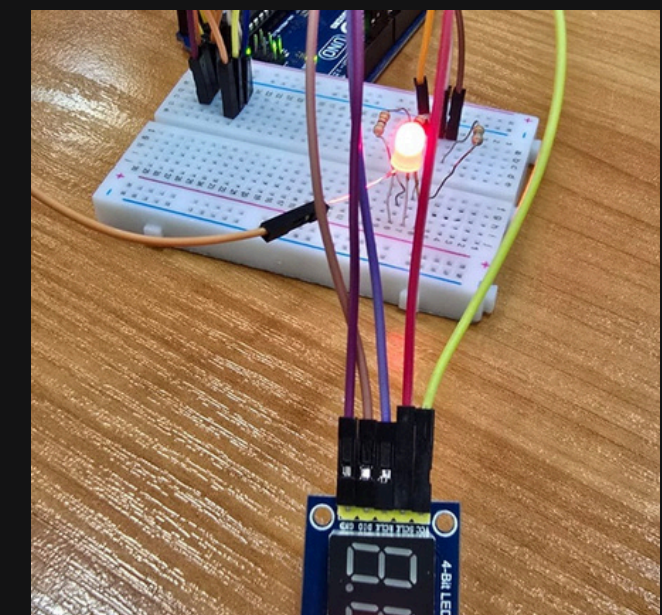
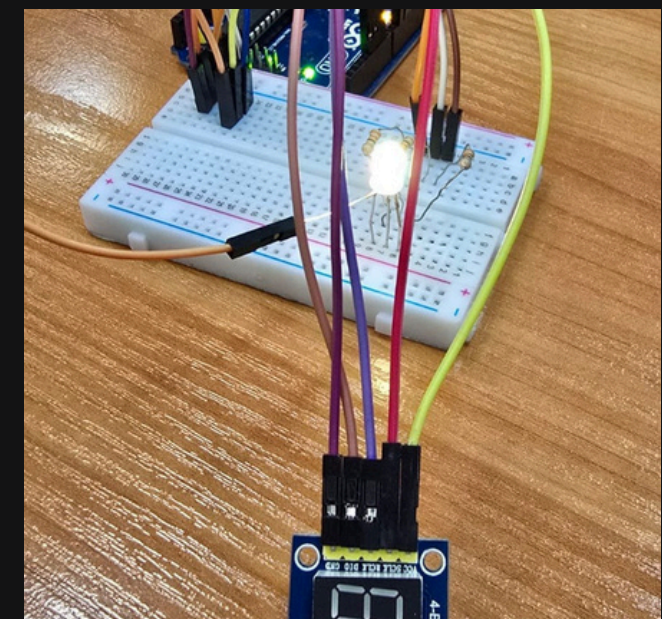
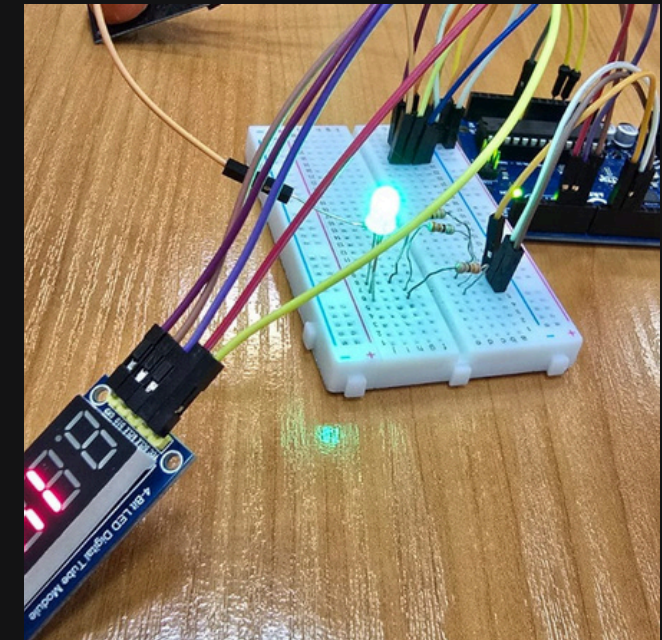
- Built a fully functional digital detector using Arduino & MQ-3.

Validation:

- The sanitizer test confirmed that the "Traffic Light" logic functions correctly.

Limitations:

- Warm-up: The sensor requires thermal stabilization before accurate readings.
- Environment: Sensitivity to temperature requires compensation for medical precision.



Thank You For Your Attention

Any question?